Operational Training Manual & Forensic Reference Guide

Design B: Explainable Artificial Intelligence (XAI) Suite

Insider Threat Intelligence Hub • Interactive Evaluation Study

Monitored Target: User 162 (`AYG1697`) **Baseline Context:** 90 Days **Schedule:** Mon–Fri, 08:00–18:00
Neural Anomaly Cutoff: 0.0742 **Task Protocol:** Two-Alert Walkthrough **Mode:** Design B (XAI Suite)

Institutional Research Ethics Statement & Voluntary Participation Protocol

This evaluation study operates in strict compliance with university-approved Office of Research Ethics (ORE) protocols. **Participation is completely voluntary.** You reserve the right to decline answering any question, skip subjective prompts, or discontinue participation at any stage without consequence.
To accommodate this autonomy within the automated database logging framework, participants choosing to pass may utilize standard bypass entries:

- **Dropdown Selections:** Select **"Prefer not to answer"** or **"Pass"** across day-level dispositions and final case closure classifications.
- **Text Input Fields:** Enter **"Pass"** or **"N/A"** in the Analyst Case Justification Narrative and interface reflection text boxes.
- **Workload Reflection (NASA-TLX):** Leave the Mental Demand and Frustration Level sliders at their default midpoint values (11).

1. Workspace Architecture & XAI Component Integration

Design B preserves all operational components from Design A (top-bar metadata, anomaly cutoff line at 0.0742, anomaly score timeline, detected anomalies table, autoencoder input features, raw user logs, and case assessment forms) while integrating a dedicated **Explainability Panel** within the Master Triage workspace.

- **Monitoring / Test Set Beeswarm Map (Global):** Summarizes global feature impact across the active monitoring period (the test set period following the initial 90-day baseline).
- **SHAP Waterfall Plots (Local):** Visualizes local feature attribution for each individual flagged alert date through interactive thumbnail cards.
- **Interactive Lightbox Viewer:** Clicking any Beeswarm or Waterfall thumbnail launches an expanded, high-resolution modal. This allows analysts to pan, zoom, inspect exact numeric SHAP contribution values, and review baseline distributions in the lateral panel.
- **AI Explainability Assistant:** An optional interactive natural language query box. Analysts can ask clarifying questions regarding feature directions (e.g., *"Which features have the highest red contributions?"*). *Operational Boundary:* The AI assistant clarifies visualizations and attributions; it does not determine maliciousness, classify scenarios, or make decisions for you.

2. Methodological Synergy: SHAP Contributions vs. Z-Scores

SHAP visual attributions and statistical Z-scores address distinct, complementary aspects of behavioral analysis. Neither metric establishes malicious intent in isolation.

Analytical Metric	Core Diagnostic Question	Operational Interpretation
Feature-Level Z-Score	*"How statistically rare is this specific feature compared to baseline?"*	Measures standard deviations (σ) from baseline mean. Values $> 3.0\sigma$ flag severe statistical outliers.
Local SHAP Attribution (Waterfall Plot)	*"In what direction and with what force did this feature push the model score?"*	Red bars push the anomaly score higher (above baseline expectation); blue bars push it lower. Read top to bottom.
Global SHAP Attribution (Beeswarm Map)	*"Which behavioral features globally drive anomalies across the monitoring window?"*	Ranks overall feature importance top-to-bottom. Horizontal spread depicts score push; dot color represents relative feature magnitude.

Critical Analytic Rules for XAI Interpretation

- **Color Signifies Score Direction, Not Maliciousness:** In Waterfall plots, red bars increase the explained anomaly score; blue bars decrease it. Red does not inherently mean "threat," and blue does not inherently mean "benign."
- **Near-Zero Values Do Not Disprove Activity:** A near-zero SHAP value does not confirm an activity never took place; it only indicates the feature exerted minimal relative leverage on that specific score.
- **Global Presence $\neq$ Temporal Co-occurrence:** Features appearing together in the global Beeswarm map do not necessarily occur on the same day. All candidate patterns must be corroborated locally using the Waterfall plot and verified chronologically in the Raw User Actions log.

3. Structured Threat Scenario Checklist

Use the **Hierarchical Triage Principle**: verify the mandatory Primary Indicator first. If absent, the scenario is dismissed. If present, cross-reference designated Secondary Indicators and corroborate using raw action timestamps.

Code	Scenario Name	Primary Indicator (Mandatory)	Secondary Indicators (Corroborating)
S-1	IP Theft via Media	**Wikileaks Web Access**	Thumb Drive Usage, After-Hours Logon
S-2	Fraud via Job & USB	**Job Site Browsing**	Competitor Email Traffic, Thumb Drive Usage
S-3	Misuse via Keylogger	**Keylogger Software Download**	Unauthorized Host Login, Mass External Email
S-4	Data Exfiltration	**Unauthorized Host Login**	Deep File System Search, Personal Webmail
S-5	Mixed Profile	**Overlapping Multi-Vector Spikes**	Synchronous cross-vector volume anomalies
S-6	Benign Anomaly	**Isolated Non-Threat Deviation**	Workload spikes, safe shift changes; no exfiltration

4. Design B: 8-Step SHAP-Assisted Investigation Workflow

1. **Select Flagged Date:** Click the anomalous date row in the *Detected Anomalies* table.
2. **Global Feature Prioritization:** Inspect the *Beeswarm Map* to assess which features generally drive anomalies across the monitoring window.
3. **Local SHAP Attribution:** Open the target date's *Waterfall Plot* in the Lightbox Viewer. Scan top-down to identify the primary red (positive) contributors.
4. **Consult AI Support (Optional):** Query the AI assistant if complex feature directions, magnitudes, or attributions require clarification.

5. **Scenario Checklist Screening:** Match prominent red-contributing features against the mandatory primary indicators in Scenarios S-1 through S-6.
6. **Empirical Feature Verification:** Examine the *Autoencoder Input Features* table and activate feature Z-scores (2σ–3σ, $> 3\sigma$) to verify physical action counts.
7. **Chronological Raw Log Audit:** Open *Raw User Actions* to verify the chronological timeline, web addresses, file access events, and device connections.
8. **Record Day-Level Disposition:** Log the day-level verdict (`Confirm Threat`, `Dismiss Alarm`, or `Prefer not to answer`) and assign the appropriate *Day Scenario* code.

5. Guided Alert Walkthroughs

5.1 Training Alert 1: October 13, 2010 (Dismiss Alarm / Benign Anomaly)

- **Global Beeswarm Review:** Identifies global candidate behaviors across the monitoring period, providing an initial baseline for investigation.
- **Local Waterfall Attribution:** In the Lightbox Viewer, the primary positive (red) contributors increasing the anomaly score are elevated *Job Search Web Browsing* and *External Email Volume*.
- **Scenario Checklist Matching:** Job portal activity activates **S-2 (Fraud via Job & USB)** as an active candidate. Primary indicators for S-1, S-3, and S-4 are completely absent.
- **Secondary Indicator & Feature Check:** Autoencoder features verify competitor email traffic; however, **Thumb Drive activity is completely absent** (0). Removable media does not appear among contributing SHAP features.
- **Raw Action Log Verification:** System events occur strictly within normal working hours (08 : 00–18 : 00). Logs confirm job site browsing but show no file staging, unauthorized access, or data egress.
- **Diagnostic Synthesis:** SHAP highlights isolated job-search behavior. The lack of thumb drive usage and exfiltration confirms this is an isolated, non-malicious deviation.
- **Logged Decision:**
  - **Operational Disposition:** `Dismiss Alarm`
  - **Day Scenario:** `S-6 (Benign Anomaly)`

5.2 Training Alert 2: October 26, 2010 (Confirm Threat / IP Theft via Media)

- **Global Beeswarm Review:** Corroborates that removable media and after-hours activities represent significant macro-level drivers across the monitoring window.
- **Local Waterfall Attribution:** Opening the October 26 Lightbox reveals dominant red bars pushing up the anomaly score: *Night-Time Login Hours*, *Elevated File Access Count*, *Thumb Drive Usage*, and *Web Access to wikileaks.org*.
- **Scenario Checklist Matching:** Outbound web requests to `wikileaks.org` satisfy the mandatory primary indicator for **S-1 (IP Theft via Media)**.
- **Secondary Indicator & Feature Check:** The Waterfall plot and Autoencoder table confirm concurrent extreme deviations in both designated secondary indicators: *Thumb Drive Usage Count* and *After-Hours Activity Count*.
- **Raw Action Log Verification:** Shortly after midnight, the event stream documents sequential access to sensitive internal files, an outbound connection to `wikileaks.org`, a USB drive disconnection, and a session logoff.
- **Diagnostic Synthesis:** SHAP pinpoints the primary and secondary contributors, which are corroborated by an unbroken, chronological multi-step exfiltration sequence in the raw logs.
- **Logged Decision:**

- **Operational Disposition:** `Confirm Threat`
- **Day Scenario:** `S-1 (IP Theft via Media)`

6. Case Assessment & Submission Protocol

After independently evaluating all flagged dates in the timeline:

1. **Verify Day-Level Completeness:** Ensure every row in the *Detected Anomalies* table has an assigned disposition. *(For this demonstration, uninspected dates may be marked Dismiss Alarm / Pass as non-binding placeholders).*
2. **Overall Case Closure Classification:** Assign the definitive case disposition for User 162 (**S-1: IP Theft via Media**, based on the corroborated October 26 threat sequence).
3. **Analyst Case Justification Narrative:** Summarize how the Beeswarm and Waterfall plots directed your diagnostic focus, which primary and secondary indicators were substantiated, and how the raw event logs confirmed or refuted the threat.
4. **Analyst Metadata:** Provide your assigned **SOC Analyst ID** and record your **Analyst Confidence Level** (1–5, or *Pass*).
5. **Subjective Reflections:** Complete the workload scales (Mental Demand, Frustration Level) and the Design B Interface Reflection prompts.
6. **Evidence Export:** Click **"Export Forensic Evidence Package"**. This securely compiles your telemetry, row verdicts, and narrative justification into the research dataset.